

Midterm 1/2022 – Part 2.1

1. Given the information below, draw grant charts for the algorithms. Also calculate average waiting time and average turnaround time for each algorithm. (12 pts)

Process Arrival Time Burst Time

P1 0.0 6
P2 2.0 3
P3 3.0 1
P4 4.0 4
P5 6.0 2

- a. FCFS (non-preemptive)



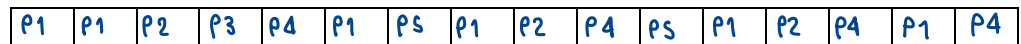
Average waiting time 4.8 Average Turnaround time 7.8

- b. SJF (preemptive)



Average waiting time 0.8 Average Turnaround time 6.2

- c. RR (quantum = 1)



Average waiting time 0 Average Turnaround time 8

- d. Which algorithm has

- Smallest average waiting time RR
- Largest average waiting time FCFS
- Smallest average turnaround time SFJ
- Largest average turnaround time RR

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Midterm 1/2022 – Part 2.2

- (13 points) Draw three tables show the 3 Frames in physical memory using FIFO, LRU and OPT for page replacement policy (separate table for both policies) and find the Effective Access Time (EAT) of each policies when references strings (list of needed pages) as below:

7, 0, 1, 3, 0, 3, 2, 4, 2, 7, 0, 3, 2, 1, 2, 0, 1, 4, 0, 1 (20 needed pages)

Assume that the memory access time is 10 ns and the disk access time is 10 ms.

FIFO

7	0	1	3	0	3	2	4	2	7	0	3	2	1	2	0	1	4	0	1
7	7	7	3	3	3	3	3	3	7	7	7	2	2	2	2	2	4	4	4
	0	0	0	0	0	2	2	2	2	0	0	0	1	1	1	1	1	1	1
		1	1	1	1	1	4	4	4	4	3	3	3	3	0	0	0	0	0

M M M M H H M M H M M M M M M H M H M H H
HIT 7/20

$$EAT = (0.35)(10^{-9}) + (0.65)(10^{-3})$$

LRU

7	0	1	3	0	3	2	4	2	7	0	3	2	1	2	0	1	4	0	1
7	7	7	3	3	3	3	3	3	7	7	7	2	2	2	2	2	4	4	4
	0	0	0	0	0	0	4	4	4	0	0	0	1	1	1	1	1	1	1
		1	1	1	1	2	2	2	2	2	3	3	3	3	0	0	0	0	0

M M M M H H M M H M M M M M M H M H M M H
HIT = 7/20

$$EAT = (0.35)(10^{-9}) + (0.65)(10^{-3})$$

OPTIMAL

7	0	1	3	0	3	2	4	2	7	0	3	2	1	2	0	1	4	0	1
7	7	7	7	7	7	7	7	7	7	0	0	0	0	0	0	0	0	0	0
	0	0	0	0	0	0	4	4	4	4	3	3	1	1	1	1	1	1	1
		1	3	3	3	2	2	2	2	2	2	2	2	2	2	2	4	4	4

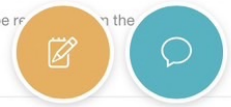
M M M M H M M M M H M M M H M H M H M H
HIT = 10/20

$$EAT = 0.5(10^{-9}) + 0.5(10^{-3})$$

ผู้จัดทำ: ภาณุ-พลเดช

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1	What is a trap in OS?	N/A	A trap is a system call that provides OS level service
2	What would happen if there is no user level and privileged level separation?	N/A	OS can prevent process from directly access devices
3	Which of the following statements is true?	N/A	All of the statements are true
4	What is not the component of a process?	N/A	All of the choices are process components
5	What is the issue to consider to protect one application from others?	N/A	Isolation
6	What is the low-level hardware that is needed in non-preemptive system to switch process?	N/A	None of them
7	Which part of the process is in kernel space?	N/A	Process descriptor
8	Which part of memory of a process grows in an opposite direction of other?	N/A	Stack
9	When would a process in ready state be moved to a blocked state?	N/A	Cannot happen
10	1. When would a process enter a ready state?	N/A	A process is created
11	What does a process in blocked state wait for?	N/A	I/O transfer to complete
12	What does it mean when process is a zombie?	N/A	Process is terminated but cannot be removed from the
13	What is the data structure that keeps	N/A	PCB



13	What is the data structure that keeps process information?	N/A	PCB
14	In the lab, when a parent waits for the child before printing, what will always happen?	N/A	Parent will always print after child
15	Which of the following scheduling algorithms execute the jobs in the arrival order?	N/A	Both FIFO and RR
16	Which event can cause a process to release from a CPU?	N/A	All of the events
17	What is not true about MLFQ?	N/A	Process enters with lowest level of priority
18	What is not true about memory?	N/A	In multitasking systems, we need physical memory that is larger than virtual memory
19	Which of the storage class is the fastest?	N/A	Register
20	Which statement is true?	N/A	Fixed-size partition suffers from internal fragmentation
21	What does process selection algorithm do?	N/A	Select process to put to memory
22	What does page replacement algorithm do?	N/A	Select a page to replace a process
23	What is not used in address translation?	N/A	All of them are used in address translation
24	What is the disadvantage of segmentation with the contiguous memory allocation?	N/A	Use too much memory to keep map of the address translation
25	Which statement is true about virtual memory in windows and Linux?	N/A	Windows system only transfer one page at a page fault
26	What does demand paging mean?	N/A	Page will be loaded when it is accessed



27	What does not happen when a page fault occurs?	N/A	All events happen
28	1. Which statement is not true?	N/A	Windows 10 is using segmentation
29	What is the data stored in the segment table?	N/A	All choices are a part of segment table
30	What is the effect of a larger page size?	N/A	Smaller page table
31	Which situation could lead to zombies in the system?	N/A	Parent exists before child returns value
32	Which process scheduling has the highest throughput?	N/A	RR
33	Which scheduler is the fairest?	N/A	RR
34	Which scheduler does not need timer interrupt?	N/A	FIFO
35	Where does virtual memory save the image of memory?	N/A	On secondary storage
36	What is the response time?	N/A	Time that process enters system until running
37	What is the turnaround time?	N/A	Time that process enters system until finish
38	What is the problem of flexible partition size?	N/A	Need compacting
39	What is the problem with paging when address space is larger?	N/A	Need more memory to store page table
40	Why multi-level page table could save some memory?	N/A	Table size is reduced
41	Which state could a process be	N/A	ready



40	Why multi-level page table could save some memory?	N/A	Table size is reduced
41	Which state could a process be when it is newly created in OS?	N/A	ready
42	When a process is using a processor, which state could it be?	N/A	running
43	What is a process in a blocked state waiting for?	N/A	Input data from disk
44	What is a disadvantage of shortest job first (SJF) scheduling?	N/A	Starvation
45	What is a disadvantage of shortest job first (SJF) scheduling?	N/A	Starvation
46	What is the advantage of round robin (RR) process scheduling?	N/A	Lowest average response time
47	What is not true about segmentation memory scheme?	N/A	All the given answer are true
48	What is true about paging memory scheme	N/A	Demand paging is when a program can demand preloading of a page
49	Which memory replacement policy is not really achieved?	N/A	Optimal
50	When does a page fault occur	N/A	When a process access a page that is not valid in page table
51	When do you want to use round robin (RR) process scheduling?	N/A	Round robin can be used in all the situation in the answers



1. What is a trap in OS?
 - a. A trap is a system call that provides OS level service
2. What would happen if there is no user level and privileged level separation?
 - a. Process may be able to access other process data
3. Which of the following statements is true?
 - a. Processes seem to occupy a CPU provided by modern OS
4. What is not the components of a process? Choice (Program, Data State, Process State, Control State)
 - a. All of the choices are process components
5. What is the issue to consider to protect one application from others? Choice(Performance, Isolation, Reliability, Mobility)
 - a. Isolation
6. What is the low-level hardware that is needed in non-preemptive system to switch process?
 - a. Timer interrupt
7. Which part of the process is in kernel space?
 - a. Process descriptor
8. Which part of memory of a process grows in an opposite direction of other?
 - a. Stack
9. When would a process in ready state be moved to a blocked state? Choice (When it read from memory, Both when read from file and memory)
 - a. When it read a file
10. When would a process enter a ready state? Choice (Timer interrupt occurs, I/O transfer is completed, A process is created)
 - a. All of the events
11. What does a process in blocked state wait for?
 - a. I/O transfer to complete
12. What does it mean when process is a zombie?
 - a. Process is terminated but cannot be removed from the system
13. What is the data structure that keeps process information? Choice(Stack, MMU, PCB, TLB, Queue)
 - a. PCB
14. In lab, when a parent waits for child before printing, what will always happen?
 - a. Parent will always print after child
15. Which of the following scheduling algorithms execute the jobs in the arrival order?
 - a. FIFO
16. Which event can cause a process to release from a CPU? Choice (Timer Interrupt, Calling I/O transfer, Calling Yield)
 - a. All of the events
17. What is not true about MLFQ?
 - a. Process enters with lowest level of priority
18. What is not true about memory?
 - a. In multitasking systems, we need physical memory that is larger than virtual memory
19. Which of the storage class is the fastest? Choice(Register, Auto, Extern, Static, They are executed at the same speed by CPU)
 - a. Register
20. Which statement is true? Choice(Single partition allocation is more efficient that multiple partition allocation in terms of memory usage, Flexible-size partition does

not suffer external fragmentation, Fixed-size partition suffers from internal fragmentation, To solve internal fragmentation, OS needs to do memory coalescing and compaction)

- a. None of them is true
- 21. What does process selection algorithm do?
 - a. Select process to run
- 22. What does page replacement algorithm do?
 - a. Select a page to remove from memory
- 23. What is not used in address translation? Choice (MMU, Page Table, TLB, Segment Table)
 - a. All of them are used in address translation
- 24. What is the disadvantage of segmentation with the contiguous memory allocation?
 - a. Use too much memory to keep map of the address translation
- 25. Which statement is true about virtual memory in windows and Linux?
 - a. Linux can use swap partition and swap file
- 26. What does demand paging mean?
 - a. Page will be loaded when it is accesses only
- 27. What does not happen when page fault occurs? Choice(Save process context, Free a frame if there is no free frame, Locate and read page from disk, Update Page Table)
 - a. All events happen
- 28. Which statement is not true? Choice(Windows 10 is using segmentation, Linux is using cluster demand paging<, In segmentation we only need one pair of base and bound, In paging, page and frame can be different in size)
 - a. All of them are false
- 29. What is the data stored in segment table? Choice (Base, Bound, Direction, Valid)
 - a. All choices are a part of segment table
- 30. What is the effect of larger page size?
 - a. Smaller page table
- 31. Which situation could lead to zombies in system?
 - a. Parent exists before child returns value
- 32. Which process scheduling has the highest throughput?
 - a. STCF
- 33. Which scheduler is the fairest?
 - a. RR
- 34. Which scheduler does not need timer interrupt?
 - a. FIFO
- 35. Where does virtual memory save the image of memory?
 - a. On secondary storage
- 36. What is the response time? Choice(Time that process starts running until finish, Time that process enters system until finish, Time that process runs until requests for I/O, Time that process runs at each cycle)
 - a. Time that process enters system until running
- 37. What is the turnaround time?
 - a. Time that process enters system until finish
- 38. What is the problem of flexible partition size? Choice(Limit degree of multitasking, Internal fragmentation, Address translation)
 - a. Need compacting
- 39. What is the problem with paging when address space is larger? Choice(Internal fragmentation, External fragmentation, TLB reach is smaller, There is no problem)
 - a. Need more memory to store page table

40. Why multi-level page table could save some memory?

- a. There is no need to store all the empty second level pages